

GIT REAL-TIME PROJECT CHEATSHEET

Practical Guide for Students – Learn Git Like Industry Professionals

1. What is Git?

Git is a Version Control System that tracks changes in your code and helps developers collaborate.

Key Purposes:

- Track code history
- Collaborate in team
- Manage versions
- Recover previous code

2. Real-Time Project Workflow

Example Project: Student Portal Application

1. Clone project
2. Create branch
3. Do changes
4. Commit
5. Pull latest code
6. Resolve conflicts
7. Push branch
8. Create Pull Request

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3. First Time Git Setup

```
git config --global user.name "Your Name"
```

```
git config --global user.email "youremail@example.com"
```

4. Starting a Project

```
git init
```

```
git clone <repo-url>
```

5. Daily Commands

```
git status
```

```
git diff
```

```
git add .
```

```
git commit -m "message"
```

6. Branching

```
git switch -c feature/login
```

```
git merge feature/login
```

7. Remote Commands

```
git push
```

```
git pull
```

8. Undo Mistakes

```
git restore filename
```

```
git reset --soft HEAD~1
```

```
git revert <commit-id>
```

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9. Stash

`git stash`

`git stash pop`

10. Tags

`git tag -a v1.0 -m "Release"`

11. Portfolio Tips

- Professional repo names
- README.md
- Regular commits

12. Must Know Commands

`git init`

`git clone`

`git status`

`git add .`

`git commit -m`

`git branch`

`git switch`

`git merge`

`git push`

`git pull`